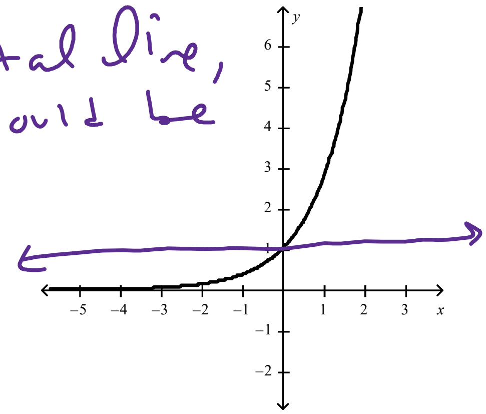
AP Calculus II
Notes 9.7
Taylor and Maclaurin Polynomials

The purpose of this next unit is to show how transcendental and trigonometric functions can be approximated by polynomials. ← $P_n(x) = ax^n + bx^{n-1} + cx^{n-2} + \dots + C$

Let's start off with trying to approximate the curve $f(x) = e^x$ at $x = 0$. The most basic approximation of this function would have a degree of 0 (a constant) and would be:

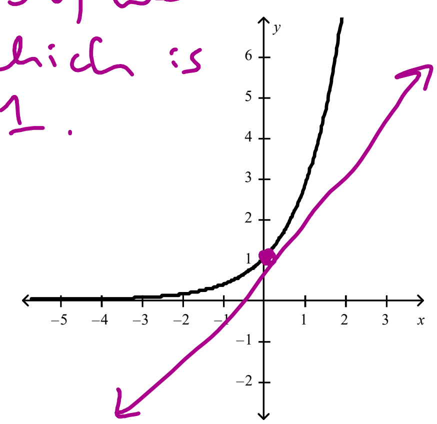
a constant would be a horizontal line, so the only thing in common would be a y-value (a point).

$P_0(x) = 1$ since $e^0 = 1$



How can we make this approximation better?

Let's add slope into the mix. So, we want point and slope to be the same, which is a tangent line! Degree is now 1.



__Point__: $(0,1)$ __slope__: $f'(x) = e^x$

$f'(0) = 1$

$y - 1 = 1(x - 0)$

or $P_1(x) = 1 + x \approx e^x$

Check:

	$P_1(x)$	$f(x) = e^x$	
point	$P_1(0) = 1$	$f(0) = e^0 = 1$	✓
slope	$P_1'(x) = 1$ $P_1'(0) = 1$	$f'(x) = e^x$ $f'(0) = 1$	✓✓

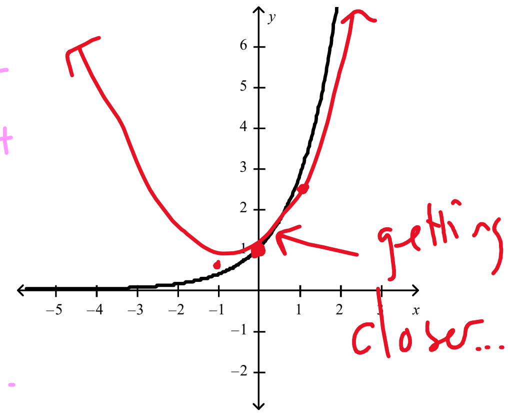
So now we have 2 things in common...

Let's add curve to the mix... Concavity!

How can we make *this* approximation any better?

increase the degree to 2!

We want same point, slope & concavity. But we aren't great at parabola writing:



$P_2(x) = 1 + x + x^2 \leftarrow$ guess...

Check:

	$P_2(x) = 1+x+x^2$	$f(x) = e^x$	
pt	$P_2(0) = 1$	$f(0) = 1$	✓
slope	$P_2'(x) = 1+2x$, $P_2'(0) = 1$	$f'(x) = e^x$, $f'(0) = 1$	✓✓
concavity	$P_2''(x) = 2$	$f''(x) = e^x$, $f''(0) = 1$	XX

So the constant and x^1 term is good, but 2^{nd} degree is twice as much... So let's divide by 2!

$P_2(x) = 1 + x + \frac{x^2}{2}$ (still guess)

	$P_2(x) = 1+x+x^2/2$	$f(x) = e^x$	
pt	$P_2(0) = 1$	$f(0) = 1$	✓
slope	$P_2'(x) = 1+x$, $P_2'(0) = 1$	$f'(x) = e^x$, $f'(0) = 1$	✓✓
concavity	$P_2''(x) = 1$, $P_2''(0) = 1$	$f''(x) = e^x$, $f''(0) = 1$	✓✓

yes!! pattern???

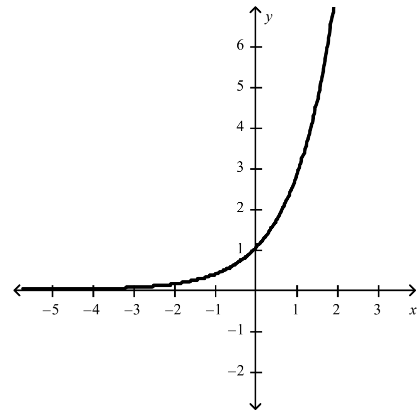
let's get to a 3rd degree and third derivative!

How can we make *this* approximation any better?

Write a 3rd degree polynomial that has the same point, slope, concavity and ... change in concavity

Guess: $P_3(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{3}$

maybe??



$P_3' = 1 + x + x^2$ $\quad f' = e^x$

$P_3'' = 1 + 2x$ $\quad f'' = e^x$

$P_3''' = 2$ $\quad f''' = e^x$

won't be same... so let's divide the $\frac{x^3}{3}$ by 2

<u>Check:</u>

$$P_3(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} \leftarrow \text{Guess}$$

$$P_3 = 1 + x + \frac{x^2}{2} + \frac{x^3}{6}\Big|_{x=0} = 1 \checkmark \qquad e^0 = 1$$

$$P_3' = 1 + x + \frac{x^2}{2}\Big|_{x=0} = 1 \checkmark\checkmark \qquad e^0 = 1$$

$$P_3'' = 1 + x\Big|_{x=0} = 1 \checkmark\checkmark\checkmark \qquad e^0 = 1$$

$$P_3''' = 1 \qquad = 1 \checkmark\checkmark\checkmark\checkmark \qquad e^0 = 1$$

So $P_3(x) = 1 + \frac{x}{1} + \frac{x^2}{2} + \frac{x^3}{6} \approx e^x$

pattern???

Definition of Maclaurin Polynomials:

nth derivative of f(x) at x=0

If $f(x)$ has n derivatives at $c = 0$, then:

change of concavity

slope

$$P_n(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \cdots + \frac{f^{(n)}(0)}{n!}x^n$$

ensures same point   concavity   and so on.....   called the general term

is also known as the **nth Maclaurin polynomial for** $f(x)$ that is centered at $c = 0$.

Ex. 1: Find the Maclaurin polynomials P_4, P_5 and P_6 for $f(x) = e^x$ and use each to approximate $e^{0.1}$.

$P_4(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} \approx e^x$ | $P_4(0.1) = 1.10517083$

$P_5(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \frac{x^5}{120} \approx e^x$ | $P_5(0.1) = 1.105170917$

$P_6(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \frac{x^5}{120} + \frac{x^6}{720} \approx e^x$ | $P_6(0.1) = 1.105170918\overline{3}$

$\boxed{e^{0.1} = 1.10517091807\overline{5}\ldots}$ accurate!

Ex. 2: Find the Maclaurin polynomial of degree 6 for $f(x) = \cos 2x$. How many nonzero terms are in the polynomial?

$P_6(x) = f(0) + \frac{f'(0)}{1!}x^1 + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \frac{f^{(6)}(0)}{6!}x^6$

we need $f(0) \longrightarrow f^{(6)}(0)$ for $f(x) = \cos 2x$

	$x=0$
$f = \cos 2x$	$\to 1$
$f' = -2\sin 2x$	$\to 0$
$f'' = -4\cos 2x$	$\to -4$
$f''' = 8\sin 2x$	$\to 0$
$f^{(4)} = 16\cos 2x$	$\to 16$
$f^{(5)} = -32\sin 2x$	$\to 0$
$f^{(6)} = -64\cos 2x$	$\to -64$

$P_6(x) = 1 + \frac{-4}{2!}x^2 + \frac{16}{4!}x^4 + \frac{-64}{6!}x^6$

$\boxed{P_6(x) = 1 - \frac{4}{2!}x^2 + \frac{16}{4!}x^4 - \frac{64}{6!}x^6 \approx \cos 2x}$

note symbols

there are 4 nonzero terms

not necessarily 4th degree

Ex. 3: Find the first 4 nonzero terms of the Maclaurin polynomial for $g(x) = \frac{6}{2+x}$.

we can assume it goes to $P_3(x)$ and can add more terms if needed:

$$P_3(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3$$

	$x=0$
$f = \frac{6}{2+x}$	$\rightarrow 3 = 3$
$f' = \frac{-6}{(2+x)^2}$	$\rightarrow -6/4 = -3/2$
$f'' = \frac{12}{(2+x)^3}$	$\rightarrow 12/8 = 3/2$
$f''' = \frac{-36}{(2+x)^4}$	$\rightarrow \frac{-36}{16} = \frac{-9}{4}$

$$\boxed{P_3(x) = 3 - \frac{3/2}{1!}x^1 + \frac{3/2}{2!}x^2 - \frac{9/4}{3!}x^3}$$

or

$$\boxed{P_3(x) = 3 - \frac{3}{2}x + \frac{3}{4}x^2 - \frac{3}{8}x^3}$$

Ex. 4: Find the Maclaurin polynomial of degree 6 for $f(x) = \cos x + \sin x$. How many nonzero terms are in the polynomial?

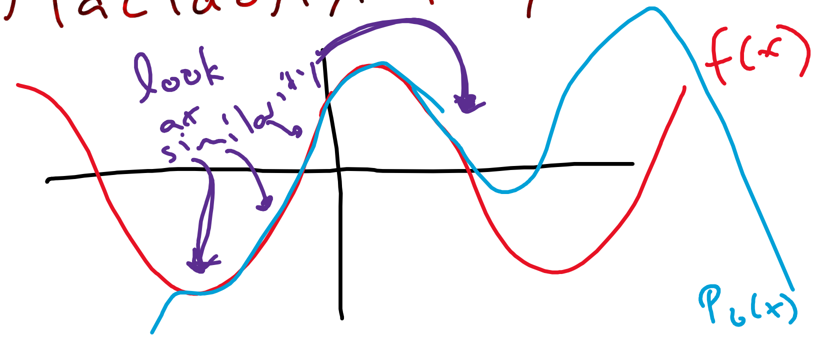
must go to 6th derivative, $6!$, x^6

$$P_6(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \frac{f^{(6)}(0)}{6!}x^6$$

	$x=0$
$f = \cos x + \sin x$	$= 1$
$f' = -\sin x + \cos x$	$= 1$
$f'' = -\cos x - \sin x$	$= -1$
$f''' = \sin x - \cos x$	$= -1$
$f^{(4)} = \cos x + \sin x$	$= 1$
$f^{(5)} = -\sin x + \cos x$	$= 1$
$f^{(6)} = -\cos x - \sin x$	$= -1$

$$\boxed{P_6(x) = 1 + \frac{1}{1!}x - \frac{1}{2!}x^2 - \frac{1}{3!}x^3 + \frac{1}{4!}x^4 + \frac{1}{5!}x^5 - \frac{1}{6!}x^6 \approx \cos x + \sin x}$$

7 terms in 6th degree Maclaurin Polynomial



what polynomial about .. starts to conform "centered" at $x = c$

Definition of Taylor Polynomials:

If at c, then:

$$P_n(x) = f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \frac{f'''(c)}{3!}(x-c)^3 + \cdots + \frac{f^{(n)}(c)}{n!}(x-c)^n$$

is also known as the **nth Taylor polynomial for** $f(x)$. A Maclaurin Polynomial is a specific type of Taylor Polynomial with $c = 0$.

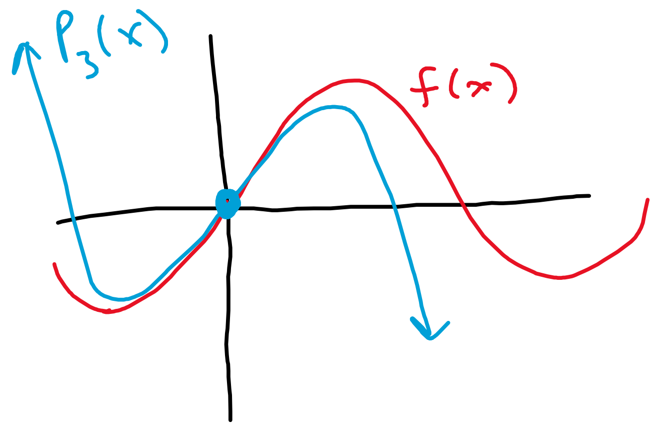
Ex. 5: Find the third-degree Taylor polynomial for $f(x) = 2\sin x$, centered at the following values:

a) $c = 0$

$$P_3(x) = f(0) + \frac{f'(0)}{1!}x^1 + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3$$

$f = 2\sin x \;\big|\; = 0$
$f' = 2\cos x \;\big|\; = 2$
$f'' = -2\sin x \;\big|\; = 0$
$f''' = -2\cos x \;\big|\; = -2$
$x = 0$

$$\boxed{P_3(x) = 2x - \frac{2}{3!}x^3}$$

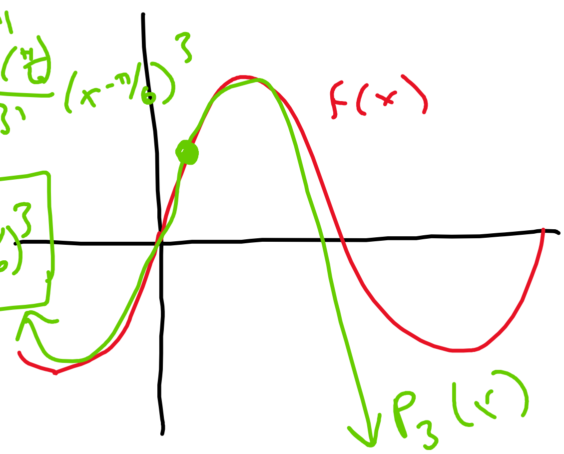


b) $c = \frac{\pi}{6}$

$$P_3(x) = f(\pi/6) + \frac{f'(\pi/6)}{1!}(x - \pi/6) + \frac{f''(\pi/6)}{2!}(x - \tfrac{\pi}{6})^2 + \frac{f'''(\pi/6)}{3!}(x - \pi/6)^3$$

$f \;\big|\; = 1$
$f' \;\big|\; = \sqrt{3}$
$f'' \;\big|\; = -1$
$f''' \;\big|\; = -\sqrt{3}$
$x = \pi/6$

$$\boxed{P_3(x) = 1 + \sqrt{3}(x - \pi/6) - \frac{1}{2!}(x - \pi/6)^2 - \frac{\sqrt{3}}{3!}(x - \pi/6)^3}$$

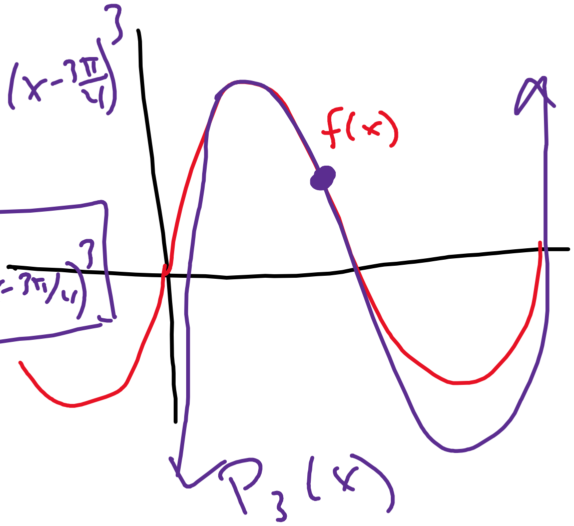


c) $c = \frac{3\pi}{4}$

$$P_3(x) = f\left(\tfrac{3\pi}{4}\right) + \frac{f'(3\pi/4)}{1!}\left(x - \tfrac{3\pi}{4}\right) + \frac{f''(\frac{3\pi}{4})}{2!}\left(x - \tfrac{3\pi}{4}\right)^2 + \frac{f'''(\frac{3\pi}{4})}{3!}\left(x - \tfrac{3\pi}{4}\right)^3$$

$f \;\big|\; = \sqrt{2}$
$f' \;\big|\; = -\sqrt{2}$
$f'' \;\big|\; = -\sqrt{2}$
$f''' \;\big|\; = \sqrt{2}$
$c = 3\pi/4$

$$\boxed{P_3(x) = \sqrt{2} - \sqrt{2}\left(x - \tfrac{3\pi}{4}\right) - \frac{\sqrt{2}}{2!}\left(x - \tfrac{3\pi}{4}\right)^2 + \frac{\sqrt{2}}{3!}\left(x - 3\pi/4\right)^3}$$



Ex. 4: Find the Taylor polynomial P_3 for $f(x) = \ln|\sin x|$ at $c = \frac{\pi}{2}$. Then, use P_3 to approximate $f(1.5)$.

$$P_3 = f(\pi/2) + \frac{f'(\pi/2)}{1!}(x-\pi/2)^1 + \frac{f''(\pi/2)}{2!}(x-\pi/2)^2 + \frac{f'''(\pi/2)}{3!}(x-\pi/2)^3$$

$f = \ln|\sin x|$ $= 0$

$f' = \frac{1}{\sin x}\cdot\cos x = \cot x$ $= 0$

$f'' = -\csc^2 x = \frac{-1}{\sin^2 x}$ $= -1$

$f''' = -2\csc x(-\csc x \cot x)$ $= 0$

$c = \pi/2$

$$\boxed{P_3(x) = \frac{-1}{2!}(x-\pi/2)^2}$$

$$\boxed{P_3(1.5) = -0.0025060599}$$

$f(1.5) = -0.0025081562$

$P_3(1.5) \approx f(1.5)$!!

Ex. 5: Find the fourth degree term of the Taylor polynomial for $f(x) = x^2 + \ln x$ centered at $c = 1$.

just term!! NOT whole polynomial!

$\frac{f^{(4)}(1)}{4!}(x-1)^4$ is 4th degree term

$f = x^2 + \ln x$

$f' = 2x + \frac{1}{x}$

$f'' = 2 - \frac{1}{x^2}$

$f''' = \frac{2}{x^3}$

$f^{(4)} = \frac{-6}{x^4}\Big|_{x=1} = -6$

$$\boxed{\frac{-6}{4!}(x-1)^4}$$

next derivative $= -2 \cdot$ previous derivative

Ex. 6: Let f be a function such that $f(2) = 4$, $f'(2) = -3$ and $f^{(n+1)}(2) = -2f^{(n)}(2)$, for all $n \geq 1$.

a) Write the first four nonzero terms of the Taylor Polynomial for f about $x = 2$.

$$P_3(x) = f(2) + \frac{f'(2)}{1!}(x-2)^1 + \frac{f''(2)}{2!}(x-2)^2 + \frac{f'''(2)}{3!}(x-2)^3$$

$f(2) = 4$
$f'(2) = -3$
$f''(2) = -2f'(2) = 6$
$f'''(2) = -12$

$$\boxed{P_3(x) = 4 - 3(x-2) + \frac{6}{2!}(x-2)^2 - \frac{12}{3!}(x-2)^3}$$

b) Write the fourth degree Taylor Polynomial for g, where $g(x) = f'(x)$ about $x = 2$.

$$P_4(x) = g(2) + \frac{g'(2)}{1!}(x-2)^1 + \frac{g''(2)}{2!}(x-2)^2 + \frac{g'''(2)}{3!}(x-2)^3 + \frac{g^{(4)}(2)}{4!}(x-2)^4$$

$g(2) = f'(2) = -3$, $g'(2) = f''(2) = 6$, $g''(2) = f'''(2) = -12$, $g'''(2) = 24$, $g^{(4)}(2) = -48$

$$\boxed{P_4(x) = -3 + 6(x-2) - \frac{12}{2!}(x-2)^2 + \frac{24}{3!}(x-2)^3 - 48/4!\,(x-2)^4}$$

c) Write the third degree Taylor Polynomial for h, where $h(x) = x^3 + \int_2^{3x-4} f(t)dt$ about $x = 2$.

$$P_3(x) = h(2) + \frac{h'(2)}{1!}(x-2) + \frac{h''(2)}{2!}(x-2)^2 + \frac{h'''(2)}{3!}(x-2)^3$$

$h(2) = 8 + \int_2^2 f(t)dt = 8$

$h'(x) = 3x^2 + f(3x-4)\cdot 3$, $h'(2) = 12 + 3f(2) = 12 + 12 = 24$

$h''(x) = 6x + 9f'(3x-4)$, $h''(2) = 12 + 9f'(2) = 12 - 27 = -15$

$h'''(x) = 6 + 27f''(3x-4)$, $h'''(2) = 6 + 27f''(2) = 6 + 162 = 168$

$$\boxed{P_3(x) = 8 + 24(x-1) - \frac{15}{2!}(x-1)^2 + \frac{168}{3!}(x-1)^3}$$

Ex. 7: The third-degree Taylor polynomial for the function f about $x=0$ is $T(x)=3-4x+2x^2-3x^3$. Which of the following tables gives the values of f and its first three derivatives at $x=0$?

(a)

x	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$
0	3	−8	6	−12

(b)

x	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$
0	3	−4	2	−3

(c)

x	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$
0	3	−4	4	−18

(d)

x	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$
0	3	−4	4	−9

$T(x)=3-4x+2x^2-3x^3$

$3=f(0) \rightarrow f(0)=3$

$-4x=\frac{f'(0)}{1!}x \rightarrow f'(0)=-4$

$2x^2=\frac{f''(0)}{2!}x^2 \rightarrow f''(0)=4$

$-3x^3=\frac{f'''(0)}{3!}x^3 \rightarrow f'''(0)=-18$

Ex. 8: The functions f, f', and f'' are each continuous and differentiable. The nth derivative of f is given by $f^{(n)}(1)=\sum_{i=0}^{\infty}12\left(\frac{n+1}{5}\right)^i$ when $0\le n\le 3$. Find the third degree Taylor polynomial for $f(x)$ centered around $x=1$.

$P_3(x)=f(1)+\frac{f'(1)}{1!}(x-1)+\frac{f''(1)}{2!}(x-1)^2+\frac{f'''(1)}{3!}(x-1)^3$

$f(1)=f^{(0)}(1)=\sum_{i=0}^{\infty}12\left(\frac{1}{5}\right)^i \xrightarrow[\text{Series!}]{\text{Geo.}} \frac{12}{1-1/5}=\frac{60}{5-1}=15$

$f'(1)=\sum_{i=0}^{\infty}12\left(\frac{2}{5}\right)^i=\frac{12}{1-2/5}=\frac{60}{5-2}=20$

$f''(1)=\sum^{\infty}12\left(\frac{3}{5}\right)^i=\frac{12}{1-3/5}=\frac{60}{5-3}=30$

$f'''(1)=\sum 12\left(\frac{4}{5}\right)^i=\frac{12}{1-4/5}=\frac{60}{5-4}=60$

$\boxed{P_3(x)=15+20(x-1)+\frac{30}{2!}(x-1)^2+\frac{60}{3!}(x-1)^3}$

AP Calculus II
Notes 9.10
Taylor and Maclaurin Power Series

In the last section, we found that Taylor and Maclaurin polynomials can approximate various functions. For example, we know the Maclaurin polynomials $P_1 \rightarrow P_5$ for the function $f(x) = e^x$ are as follows:

$P_1 = 1 + x \approx e^x$

$P_2 = 1 + x + \frac{x^2}{2!} \approx e^x$

$P_3 = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \approx e^x$

$P_4 = 1 + x + x^2/2! + x^3/3! + x^4/4! \approx e^x$

$P_5 = 1 + x + x^2/2! + x^3/3! + x^4/4! + x^5/5! \approx e^x$

We also found that the higher the degree of the polynomial, the better the approximation becomes about c. Therefore, if we wanted to find the *exact* representation of these functions, we use **power series**.

let's have ∞ many terms added together!

Definition of Power Series

If x is a variable, then an infinite series of the form:

$$\sum_{n=0}^{\infty} a_n(x-c)^n = a_0 + a_1(x-c) + a_2(x-c)^2 + a_3(x-c)^3 + \cdots + a_n(x-c)^n$$

is called a **power series centered at c** and $a_n(x-c)^n$ is called the **general term**.

Taylor and Maclaurin Power Series

If a function $f(x)$ has derivatives of all orders at $x = c$, then the series:

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!}(x-c)^n = f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \frac{f'''(c)}{3!}(x-c)^3 + \cdots + \frac{f^{(n)}(c)}{n!}(x-c)^n$$

is called the **Taylor series for $f(x)$ centered at c**. Again, a Taylor series centered at $c = 0$ is called a **Maclaurin series.**

this is the general term!

If you can find the pattern for the coefficients of the Taylor polynomials for a function, then you can extend the pattern easily for the corresponding Taylor series.

Ex. 1: Write the fourth-degree Maclaurin polynomial for $f(x) = e^x$, write the general term and the Maclaurin series for $f(x) = e^x$.

center is 0

$$P_4(x) = \frac{1}{0!}x^0 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} \approx e^x$$

the pattern seems to be $\frac{1x^n}{n!}$ same power & factorial

General term $\rightarrow \boxed{\frac{x^n}{n!}}$

$\boxed{\text{wow!}}$

∞ many terms

Maclaurin Series: $\displaystyle\sum_{n=0}^{\infty} \frac{x^n}{n!} = \underset{n=0}{1} + \underset{n=1}{x} + \underset{n=2}{\frac{x^2}{2!}} + \underset{n=3}{\frac{x^3}{3!}} + \dots = e^x$

Ex. 2: Write the fourth-degree Taylor polynomial for $f(x) = \ln x$, centered about $c = 1$, write the general term and the Taylor series for $f(x) = \ln x$, centered about $c = 1$.

$$P_4(x) = f(1) + \frac{f'(1)}{1!}(x-1) + \frac{f''(1)}{2!}(x-1)^2 + \frac{f'''(1)}{3!}(x-1)^3 + \frac{f^{(4)}(1)}{4!}(x-1)^4$$

$f: \ln x$	$= 0$	
$f' = \frac{1}{x}$	$= 1$	
$f'': \frac{-1}{x^2}$	$= -1$	
$f''': 2/x^3$	$= 2$	
$f^{(4)}: \frac{-6}{x^4}$	$= -6$	
	$c=1$	

$$P_4(x) = \frac{1}{1!}(x-1)^1 - \frac{1}{2!}(x-1)^2 + \frac{2}{3!}(x-1)^3 - \frac{6}{4!}(x-1)^4$$

$$= \frac{1}{1}(x-1)^1 - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 - \frac{1}{4}(x-1)^4$$

pattern: alternates, no factorial

$$\frac{(-1)^n(x-1)^n}{n} \rightarrow \boxed{\ln x = \sum_{n=1}^{\infty} \frac{(-1)^n}{n}(x-1)^n}$$

(start at $n=1$)

Ex. 3: Write the Maclaurin series for $f(x) = \sin x$.

Write a Maclaurin polynomial to discover the pattern

$$P_5(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5$$

$f = \sin x$ | 0
$f' = \cos x$ | 1
$f'' = -\sin x$ | 0
$f''' = -\cos x$ | -1
$f^{(4)} = \sin x$ | 0
$f^{(5)} = \cos x$ | 1
$f^{(6)} = -\sin x$ | 0
$x=0$
$f^{(7)} = -\cos x = -1$

$$P_5(x) = \frac{1}{1!}x^1 - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 \left(- \frac{1}{7!}x^7 + \frac{1}{9!}x^9 \right)$$

$n=0$ $n=1$ $n=2$ $n=3$ $n=4$

if we kept going

$$\boxed{\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}}$$

Ex. 4: Write the Maclaurin series for $f(x) = \cos x$.

$f = \cos x$ | 1
$f' = -\sin x$ | 0
$f'' = -\cos x$ | -1
$f''' = \sin x$ | 0
$f^{(4)} = \cos x$ | 1
$f^{(5)} = -\sin x$ | 0
$f^{(6)} = -\cos x$ | -1
$f^{(7)} = \sin x$ | 0
$f^{(8)} = \cos x$ | 1
$x=0$

$$P_8(x) = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \frac{1}{8!}x^8$$

$n=0$ $n=1$ $n=2$ $n=3$ $n=4$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

* Note: $2n! \neq (2n)!$

expanded form $\downarrow$ sigma form $\downarrow$

The Maclaurin Series You MUST Know:

$e^0 = 1+0+0+0 = 1$ 1) $e^x = 1 + x + x^2/2! + x^3/3! + \frac{x^4}{4!} + \dots + \frac{x^n}{n!} = \sum_{n=0}^{\infty} \frac{x^n}{n!}$

$\sin 0 = 0-0+0=0$ 2) $\sin x = x - x^3/3! + x^5/5! - x^7/7! + \dots \frac{(-1)^n x^{2n+1}}{(2n+1)!} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$

$\cos 0 = 1-0+0 = 1$ 3) $\cos x = 1 - x^2/2! + \frac{x^4}{4!} - x^6/6! + x^8/8! \dots \frac{(-1)^n x^{2n}}{(2n)!} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$

$\frac{1}{1-0} = 1+0+0 = 1$ 4) $\frac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 + \dots + x^n = \sum_{n=0}^{\infty} x^n$

$\uparrow$ geometric!

Ex. 5: Find the Maclaurin power series for $f(x) = \sin(x^2)$.

$f = \sin x^2$ | 0

$f' = 2x\cos x^2$ | 0

$f'' = 2\cos x^2 - 4x^2 \sin x^2$ | 2

$x=0$

crazy $\downarrow$ more derivatives needed

Instead:

$\sin x = x - x^3/3! + x^5/5! + \dots \frac{(-1)^n x^{2n+1}}{(2n+1)!}$

$\sin x^2 = x^2 - \frac{(x^2)^3}{3!} + \frac{(x^2)^5}{5!} + \dots \frac{(-1)^n (x^2)^{2n+1}}{(2n+1)!}$

$$\boxed{\sin x^2 = x^2 - \frac{x^6}{3!} + \frac{x^{10}}{5!} - \dots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{4n+2}}{(2n+1)!}}$$

Ex. 6: Find the Taylor power series for $f(x) = \frac{e^{(-3x)} - 1}{3}$ centered about $x = 0$.

Let's Build $f(x)$ from e^x

$e^x = 1 + x + x^2/2! + x^3/3! + \dots + \frac{x^n}{n!}$ replace x's with $-3x$

$e^{-3x} = 1 - 3x + \frac{9x^2}{2!} - \frac{27x^3}{3!} + \dots + \frac{(-3x)^n}{n!}$ Subtract 1 from both sides

$e^{-3x} - 1 = -3x + \frac{9x^2}{2!} - \frac{27x^3}{3!} + \dots + \frac{(-3x)^n}{n!}$ $\rightarrow$ starting at $n=1$ since we removed $n=0$

$$\boxed{\frac{e^{(-3x)} - 1}{3} = -x + \frac{3x^2}{2!} - \frac{9x^3}{3!} + \dots + \frac{(-3x)^n}{n! \cdot 3} = f(x) = \sum_{n=1}^{\infty} \frac{(-3x)^n}{3n!}}$$

let's write out 4 in middle steps to be sure

Ex. 7: Find the first three nonzero terms in the Maclaurin series $g(x) = x\cos x - \sin x$.

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

no need for general term

$$x\cos x = x - \frac{x^3}{2!} + \frac{x^5}{4!} - \frac{x^7}{6!} + \dots$$

$$-\left(\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots\right)$$

only add & subtract like terms & need common denominator!

$$x\cos x - \sin x = g(x) = \left(\frac{-1}{2!} + \frac{+1}{3!}\right)x^3 + \left(\frac{1}{4!} - \frac{1}{5!}\right)x^5 + \left(\frac{-1}{6!} + \frac{+1}{7!}\right)x^7 + \dots$$

$$= -\frac{2}{3!}x^3 + \frac{4}{5!}x^5 - \frac{6}{7!}x^7 + \dots$$

Ex. 8: Evaluate the following infinite series:

a) $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n \pi^{2n}}{3^{2n}(2n)!} = \sum_{n=0}^{\infty} \frac{(-1)^n (\pi/3)^{2n}}{(2n)!}$

focus on factorial looks like cosine

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

replaces x with $\pi/3$

$$\text{So, } \sum_{n=0}^{\infty} \frac{(-1)^n \pi^{2n}}{3^{2n}(2n)!} = \cos \pi/3 = \boxed{\frac{1}{2}}$$

b) $4 - \frac{4x^2}{3!} + \frac{4x^4}{5!} - \frac{4x^6}{7!} + \cdots$

Factorials look like $\sin x$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \quad \cdot 4$$

$$4\sin x = 4x - \frac{4x^3}{3!} + \frac{4x^5}{5!} - \frac{4x^7}{7!} + \dots \quad \div x$$

$$\boxed{\frac{4\sin x}{x} = 4 - \frac{4x^2}{3!} + \frac{4x^4}{5!} - \frac{4x^6}{7!} + \dots}$$

c) $1 - 2 + \frac{4}{2!} - \frac{8}{3!} + \frac{16}{4!} - \frac{32}{5!} + \cdots$

(with $1!$ under 2 and under 1)

Factorials look like e^x

$$\text{sigma notation: } \sum_{n=0}^{\infty} \frac{(-1)^n 2^n}{n!}$$

$$\sum_{n=0}^{\infty} \frac{(-2)^n}{n!} \quad : \quad e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

So replace x with -2

$$\boxed{1 - 2 + \frac{4}{2!} - \frac{8}{3!} + \frac{16}{4!} - \frac{32}{5!} + \dots = e^{-2}}$$

d) $1 - \frac{x^4}{2!} + \frac{x^8}{4!} - \frac{x^{12}}{6!} + \cdots$

factorial looks like $\cos x$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

powers are doubled

$$\cos(x^2) = 1 - \frac{(x^2)^2}{2!} + \frac{(x^2)^4}{4!} - \frac{(x^2)^6}{6!} + \dots$$

$$\boxed{\cos(x^2)} = 1 - \frac{x^4}{2!} + \frac{x^8}{4!} - \frac{x^{12}}{6!} + \dots$$

Ex. 9: Given $f(x) = 2x - \frac{2x^3}{3!} + \frac{2x^5}{5!} - \frac{2x^7}{7!} + \cdots + \frac{(-1)^n 2x^{2n+1}}{(2n+1)!}$, ← 2 isn't raised to any power, so the 2 isn't with the x

a) Find one value of x for which $f(x) = 1$.

can't solve $1 = 2x - \frac{2x^3}{3!} + \frac{2x^5}{5!} - \frac{2x^7}{7!} + \cdots$ as polynomial

So, let's rewrite polynomial as function

$f(x) = 2\sin x$ $\quad 2\sin x = 1,\ \sin x = 1/2$ at $\boxed{x = \pi/6}$

b) Find the first 4 non-zero terms for $f'(x)$. Write the Maclaurin series for $f'(x)$?

$f'(x)$ is multiple constants and power rules

$f'(x) = 2 - \frac{6x^2}{3!} + \frac{10x^4}{5!} - \frac{14x^6}{7!} + \cdots + \frac{(-1)^n 2(2n+1)x^{2n}}{(2n+1)!}$ ← multiply power & subtract 1

$f'(x) = 2 - \frac{2x^2}{2!} + \frac{2x^4}{4!} - \frac{2x^6}{6!} + \cdots + \frac{(-1)^n 2x^{2n}}{(2n)!}$ ← divided a factor in numerator & had 1 less factorial

$\boxed{f'(x) = 2\cos x}$ $\quad \boxed{\text{if } f = 2\sin x,\ f' = 2\cos x}$

c) If $g'(x) = f(x)$ and $g(0) = -2$, find the first 4 non-zero terms and the general term of $g(x)$.

$g' = 2x - \frac{2x^3}{3!} + \frac{2x^5}{5!} - \frac{2x^7}{7!} + \cdots + \frac{(-1)^n 2x^{2n+1}}{(2n+1)!}$ integrate

$g = x^2 - \frac{2x^4}{4 \cdot 3!} + \frac{2x^6}{6 \cdot 5!} - \frac{2x^8}{8 \cdot 7!} + \cdots + \frac{(-1)^n 2x^{(2n+2)}}{(2n+2)(2n+1)!} + C$

$g = C + x^2 - \frac{2x^4}{4!} + \frac{2x^6}{6!} - \frac{2x^8}{8!} + \cdots + \frac{(-1)^n 2x^{(2n+2)}}{(2n+2)!}$

$-2 = C + 0 - 0 + 0 - 0 + \cdots \quad C = -2$ $\quad (g_{(0)} = -2)$

$\boxed{g = -2 + x^2 - \frac{2x^4}{4!} + \frac{2x^6}{6!} - \cdots \frac{(-1)^n 2x^{2n}}{(2n)!}}$

d) Find $f^{(19)}(0)$.

19th degree term is

$\frac{f^{(19)}(0)}{19!} x^{19} \rightarrow$ general term is $\frac{(-1)^n 2x^{(2n+1)}}{(2n+1)!}$

uses $n = 9$

$\frac{(-1)^9 2x^{19}}{19!} \rightarrow$ so $f^{(19)}(0) = (-1)^9 \cdot 2 = \boxed{-2}$

AP Calculus II
Notes 9.8
Power Series

In the last section, we dealt with Maclaurin and Taylor polynomials that when added up, resulted in Maclaurin and Taylor series. In this section, we will focus on any power series and its features.

Definition of Power Series

If x is a variable, then an infinite series of the form:

$$\sum_{n=0}^{\infty} a_n(x-c)^n = a_0 + a_1(x-c) + a_2(x-c)^2 + a_3(x-c)^3 + \cdots + a_n(x-c)^n$$

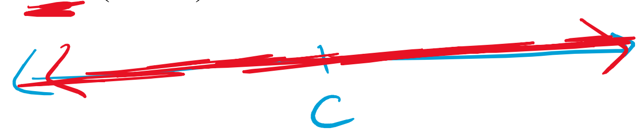
is called a **power series centered at** c.

When representing a function $f(x)$ as a power series, the **domain** of the function $f(x)$ is all the values of x for which the power series converges. The Geometric, Alternating Series and Ratio Tests are very helpful in these situations, but the other tests are just as useful. The set of all x −values for which the power series converges is known as the **interval of convergence**.

Interval of Convergence of a Power Series:

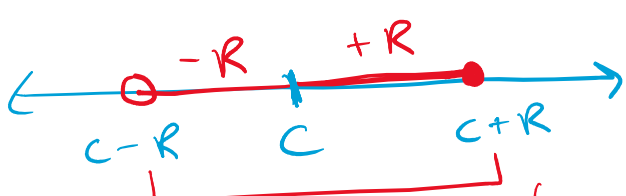
For a power series centered at c, one of the following cases is true, along with a graphical representation:

a) The series converges for all x. ($R = \infty$) radius of convergence is ∞



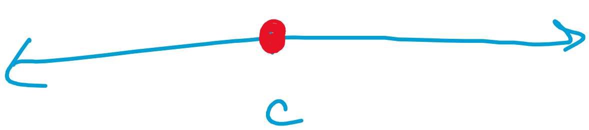
b) The series converges for only a finite window $|x-c| < R$, with c being the center of this window and R being the **radius of convergence**. Think of R as being the "plus or minus" away from the center. This interval of convergence can include one, none or both endpoints.

may be $<$ or $\leq$



only x-values that "work"

c) The series converges at one point, namely c. ($R = 0$)



only x-value that works is at $x=c$

Use Ratio or Geometric Test ... which converges if $|r|<1$

Ex. 1: Find the center, radius and interval of convergence for the following power series:

a) $\sum_{n=1}^{\infty} \frac{x^n}{n}$

$\lim_{n\to\infty} \left| \frac{x^{n+1}}{n+1} \cdot \frac{n}{x^n} \right| < 1 \rightarrow \lim_{n\to\infty} \left| x \cdot \frac{n}{n+1} \right| < 1$

as $n \to \infty$, this goes to 1

$|x| < 1 \rightarrow -1 < x < 1$

center is $\boxed{0}$ since $x - 0$

$c=0$

-1 $+1$

So radius is $\boxed{1}$

Since there is a constant radius, we must test the endpoints

$x=-1$: $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ which $\boxed{\text{converges}}$ because it's alternating harmonic

$x=1$: $\sum_{n=1}^{\infty} \frac{(1)^n}{n} = \sum_{n=1}^{\infty} \frac{1}{n}$ which is $\boxed{\text{divergent}}$ harmonic

So, $\boxed{-1 \le x < 1}$

b) $\sum_{n=1}^{\infty} n!\,x^n$

$\lim_{n\to\infty} \left| \frac{(n+1)!\,x^{n+1}}{n!\,x^n} \right| < 1$, $\lim_{n\to\infty} \left| (n+1) \cdot x \right| < 1$

as $n \to \infty$, $n+1 \to \infty$, which is never less than 1.

$\boxed{\begin{array}{l} \text{Center}: c = 0 \\ \text{radius}: R = 0 \end{array}}$

So the only value that would cause the series to converge is x = center $\boxed{x=0}$

c) $\sum_{n=3}^{\infty} (-1)^{n+1} \frac{(x+3)^n}{(n-2)^3}$

$\lim_{n\to\infty} \left| \frac{(x+3)^{n+1}}{(n-1)^3} \cdot \frac{(n-2)^3}{(x+3)^n} \right| < 1 \rightarrow \lim_{n\to\infty} \left| (x+3) \frac{(n-2)^3}{(n-1)^3} \right| < 1$

as $n \to \infty$, this goes to 1

$|x+3| < 1$

$-1 < x+3 < 1$

$-4 < x < -2$

$\boxed{\begin{array}{l} \text{Center}: x = -3 \\ \text{radius}: R = 1 \end{array}}$

Test Endpoints: See if $x=-4$ & $x=-2$ converge

$x=-4$: $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} \cdot (-1)^n}{(n-2)^3} = \sum \frac{(-1)^{2n+1}}{(n-2)^3}$ doesn't alternate

$\sum \frac{1}{(n-2)^3}$ $\boxed{\text{Converges}}$ to compare to $p=3$

$x=-2$: $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{(n-2)^3}$ $\boxed{\text{Converges}}$ by alternating series $\boxed{[-4, -2]}$

Ex. 2: Write out the first 4 nonzero terms and the general term of the Maclaurin series for $f(x) = \sin x$. Then, determine the interval of convergence for this function.

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots + \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

$$\lim_{n\to\infty}\left|\frac{x^{2n+3}}{(2n+3)!}\cdot\frac{(2n+1)!}{x^{2n+1}}\right| < 1 \rightarrow \lim_{n\to\infty}\left|x^2\cdot\frac{1}{(2n+3)(2n+2)}\right| < 1$$

as $n\to\infty$, this goes to 0

$|x^2\cdot 0| < 1 \rightarrow 0 < 1$ which is always true, so

$\boxed{\text{Interval of convergence}: (-\infty,\infty)}$

Ex. 3: The Taylor series for the function $f(x) = \ln x$ is $\sum_{n=1}^{\infty}\frac{(-1)^{n-1}(x-1)^n}{n}$. Find the interval of convergence for this function. Then, find the value of $\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n}$.

IoC: $\lim_{n\to\infty}\left|\frac{(x-1)^{n+1}}{n+1}\cdot\frac{n}{(x-1)^n}\right| < 1 \rightarrow \lim_{n\to\infty}\left|(x-1)\cdot\frac{n}{n+1}\right| < 1$

as $n\to\infty$, this goes to 1

$|x-1| < 1$

$-1 < x-1 < 1$

$0 < x < 2$

$x=0$: $\sum^{\infty}\frac{(-1)^{n-1}(-1)^n}{n}$ ← not alternating harmonic → Diverges

$x=2$: $\sum^{\infty}\frac{(-1)^{n-1}\cancel{(1)^n}}{n}$ ← alternating harmonic → Converges

$\boxed{\text{Ioc}: (0,2]}$

$\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n} \rightarrow \sum_{n=1}^{\infty}\frac{(-1)^{n-1}(x-1)^n}{n}$ means $(\cancel{x}-1)^n = 1$ which is when $x=2$

So, since $x=2$ is in Ioc $\rightarrow \ln x = \sum^{\infty}\frac{(-1)^{n-1}(x-1)^n}{n}$

$\boxed{\ln 2 = \sum^{\infty}\frac{(-1)^{n-1}}{n}}$ the alternating harmonic series adds up to $\ln 2$

center is $x=2$ one of the endpoints so $R=2$

Ex. 4: The series $\sum_{n=1}^{\infty} a_n (x-2)^n$ converges conditionally when $x=0$. Determine if the following statements are **A**lways true, **S**ometimes true or **N**ever true.

a.) The series diverges when $x = 5$. $\boxed{\text{Always true}}$ since beyond $R=2$	**b.)** The series converges when $x = -1$. $\boxed{\text{Never true}}$ since before $x=0$
c.) The series diverges when $x = 1$. $\boxed{\text{Never true}}$ Inside Interval of convergence	**d.)** The series converges conditionally when $x = 4$. $\boxed{\text{Sometimes true}}$ the other endpoint so either converges or diverges

Ex. 5: $\sum_{n=1}^{\infty} b_n = -\frac{4}{3} + \frac{4\cdot 2}{9}(x-2) - \frac{4\cdot 3}{27}(x-2)^3 + \cdots + (-1)^n \frac{4n}{3^n}(x-2)^{n-1} + \cdots$

Find the interval of convergence for the series $\sum_{n=1}^{\infty} \frac{b_n}{n^2}$. $= \sum \frac{(-1)^n 4\cancel{n}(x-2)^{n-1}}{3^n n^{\cancel{2}}}$

$$= \sum_{n=1}^{\infty} \frac{(-1)^n 4 (x-2)^{n-1}}{n 3^n}$$

IOC: $\lim_{n\to\infty} \left| \frac{\cancel{4}(x-2)^{\cancel{n}}}{(n+1)3^{\cancel{n+1}}} \cdot \frac{n 3^{\cancel{n}}}{\cancel{4}\cancel{(x-2)^{n-1}}} \right| < 1 \rightarrow \lim_{n\to\infty} \left| \frac{x-2}{3} \cdot \frac{n}{n+1} \right| < 1$

as $n\to\infty$, this goes to 1

$$\left|\frac{x-2}{3}\right| < 1$$

$$-1 < \frac{x-2}{3} < 1$$

$$-3 < x-2 < 3$$

$$-1 < x < 5$$

$x=-1$: $\sum \frac{(-1)^n 4(-3)^{n-1}}{n 3^n} = \sum \frac{(-1)^n 4 (3)^{n-1}(-1)^{n-1}}{n 3^n}$ not alternating!

$= \sum \frac{4}{3n}$ is $\boxed{\text{divergent}}$ harmonic

$x=5$: $\sum \frac{(-1)^n 4 (3)^{n-1}}{n 3^n} = \sum \frac{(-1)^n \cdot 4}{3n}$ is alternating harmonic, so $\boxed{\text{converges}}$

$$\boxed{-1 < x \leq 5}$$

Power Series Manipulation

Since these series represent different functions, then we can manipulate the series the same way we manipulate functions. This allows us to differentiate, integrate, find extrema, points of inflection, etc…

Ex. 6: Given $\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n!}$, find the following:

a) The first four terms and the general term for $f'(x)$. Then, find $f'(0)$ and $f^{(11)}(0)$.

$$f = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \dots + \frac{(-1)^n x^n}{n!}$$

$$f' = -1 + \frac{2x}{2!} - \frac{3x^2}{3!} + \frac{4x^3}{4!} - \dots + \frac{(-1)^n n x^{n-1}}{n!}$$

$$\boxed{f' = -1 + x - \frac{x^2}{2!} + \frac{x^3}{3!} - \dots + \frac{(-1)^n x^{n-1}}{(n-1)!}}$$

$\boxed{f'(0) = -1}$

using $n = 12 \dots$

$+\frac{x^{11}}{11!} \rightarrow \boxed{f^{(11)}(0) = 1}$

b) The first four terms and the general term for $\int f(x)dx$.

$$\int \left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots + \frac{(-1)^n x^n}{n!}\right) dx$$

$$= x - \frac{x^2}{2} + \frac{x^3}{3 \cdot 2!} - \frac{x^4}{4 \cdot 3!} + \dots + \frac{(-1)^n x^{n+1}}{(n+1)n!} + C$$

$$= \boxed{C + x - \frac{x^2}{2!} + \frac{x^3}{3!} - \frac{x^4}{4!} + \dots + \frac{(-1)^n x^{n+1}}{(n+1)!}}$$

c) The first four terms and the general term for $g'(t)$ given $g(t) = f(t^3)$.

$$f(x) = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \dots + \frac{(-1)^n x^n}{n!}$$

replace x with t^3

$$f(t^3) = 1 - t^3 + \frac{t^6}{2!} - \frac{t^9}{3!} + \frac{t^{12}}{4!} - \dots + \frac{(-1)^n t^{3n}}{n!} = g(t)$$

$$g'(t) = 0 - 3t^2 + \frac{6t^5}{2!} - \frac{9t^8}{3!} + \frac{12t^{11}}{4!} - \dots + \frac{(-1)^n 3n(t)^{3n-1}}{n!}$$

$$\boxed{g' = -3t^2 + \frac{3t^5}{1!} - \frac{3t^8}{2!} + \frac{3t^{11}}{3!} - \dots + \frac{(-1)^n 3t^{3n-1}}{(n-1)!}}$$

Constant

x to a power

Ex. 7: Given $f(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}(x-1)^{n+1}}{(n+1)(n+2)}$, find the interval of convergence for $f'(x)$.

constants

chain rule!

$$f'(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}\cancel{(n+1)}(x-1)^{n}(1)}{\cancel{(n+1)}(n+2)} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}(x-1)^{n}}{n+2}$$

IoC: $\lim_{n\to\infty} \left| \frac{(x-1)^{n+1}}{n+3} \cdot \frac{n+2}{(x-1)^n} \right| < 1 \to \lim_{n\to\infty} \left| (x-1) \cdot \frac{n+2}{n+3} \right| < 1$

as $n \to \infty$, this goes to 1

$|x-1| < 1$

$-1 < x-1 < 1$

$0 < x < 2$

$x=0$: $\sum \frac{(-1)^{n+1}(-1)^n}{n+2}$ ← not alternating harmonic, so $\boxed{\text{diverges}}$

$x=2$: $\sum \frac{(-1)^{n+1}(1)^n}{n+2}$ ← alternating harmonic, so $\boxed{\text{converges}}$

$\boxed{(0,2]}$

Ex. 8: Find the first 3 nonzero terms and the general term for $\int_0^x \sin(t^4)dt$.

u-sub doesn't work....

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots + \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

$(t^4)^{2n+1}$

$$\sin t^4 = t^4 - \frac{t^{12}}{3!} + \frac{t^{20}}{5!} - \frac{t^{28}}{7!} + \dots + \frac{(-1)^n t^{8n+4}}{(2n+1)!}$$

$$\int_0^x \sin t^4 dt = \int_0^x \left[t^4 - \frac{1}{3!}t^{12} + \frac{1}{5!}t^{20} - \frac{1}{7!}t^{28} + \dots + \frac{(-1)^n}{(2n+1)!}t^{8n+4} \right] dt$$

can't integrate

definitely can integrate!

$$\int_0^x \sin t^4 dt = \frac{1}{5}t^5 - \frac{1}{3!}\cdot\frac{t^{13}}{13} + \frac{1}{5!}\cdot\frac{t^{21}}{21} - \frac{1}{7!}\cdot\frac{t^{29}}{29} + \dots + \frac{(-1)^n}{(2n+1)!}\cdot\frac{t^{8n+5}}{(8n+5)}\Bigg|_0^x$$

$$\boxed{= \frac{1}{5}x^5 - \frac{1}{13\cdot 3!}x^{13} + \frac{1}{21\cdot 5!}x^{21} - \dots + \frac{(-1)^n}{(8n+5)(2n+1)!}\cdot x^{8n+5} - (0-0+0\dots)}$$

1 2 3 terms

Ex. 9: Given $f(x) = \frac{\cos x - 1}{x^2}$, find the following using series:

a) $\lim_{x \to 0} f(x)$, then verify using L'Hoptial's rule.

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \quad / \quad \cos x - 1 = -\frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$\frac{\cos x - 1}{x^2} = \frac{-1}{2!} + \frac{x^2}{4!} - \frac{x^4}{6!} + \dots = f(x)$$

$$\lim_{x \to 0} \frac{\cos x - 1}{x^2} = \frac{-1}{2!} + \frac{0^2}{4!} - \frac{0^4}{6!} + \dots = -\frac{1}{2} + 0 - 0 + \dots = \boxed{-\frac{1}{2}}$$

L'HOP

$$\lim_{x \to 0} \cos x - 1 = 0, \quad \lim_{x \to 0} x^2 = 0, \quad = \lim_{x \to 0} \frac{-\sin x}{2x} \quad \lim_{x \to 0} -\sin x = 0, \quad \lim_{x \to 0} 2x = 0 \quad = \lim_{x \to 0} \frac{-\cos x}{2} = \frac{-1}{2}$$

b) First 4 nonzero terms and the general term for $\int f(x)dx$.

$$f(x) = \frac{-1}{2!} + \frac{x^2}{4!} - \frac{x^4}{6!} + \frac{x^6}{8!} - \dots + \frac{(-1)^n x^{2n-2}}{(2n)!}$$

$$\boxed{\int f(x)dx = C + \frac{-1}{2!}x^1 + \frac{x^3}{3 \cdot 4!} - \frac{x^5}{5 \cdot 6!} + \frac{x^7}{7 \cdot 8!} + \dots + \frac{(-1)^n x^{2n-1}}{(2n-1)(2n)!}}$$

Ex. 10: Given $g(x) = \frac{e^x - e^{-x}}{2}$, find:

a) The power series expansion of $g(x)$ by showing the first 4 nonzero terms and general term.

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} \cdots + \frac{x^n}{n!}$$

$$-\left(e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \frac{x^5}{5!} \cdots + \frac{(-x)^n}{n!} \leftarrow \frac{(-1)^n x^n}{n!}\right)$$

$$e^x - e^{-x} = 2x + \frac{2x^3}{3!} + \frac{2x^5}{5!} + \frac{2x^7}{7!} + \cdots + \frac{2x^{2n+1}}{(2n+1)!}$$

$$\boxed{g(x) = \frac{e^x - e^{-x}}{2} = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \cdots + \frac{x^{2n+1}}{(2n+1)!}}$$

b) The relationship between $g(x)$ and the series $h(x) = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \cdots$

$$g' = 1 + \frac{3x^2}{3!} + \frac{5x^4}{5!} + \frac{7x^6}{7!} + \cdots$$

$$\boxed{g' = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \cdots = h(x)}$$

c) The equation represented in the power series expansion for the function $h(x)$.

Since $h = g'$ according to the series expansion, $h = g'$ in the equation of the function

$$\boxed{h = g' = \frac{1}{2}(e^x + e^{-x})}$$

Ex. 11:

The function f is defined by the power series

$$f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n+1)!} = 1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \frac{x^6}{7!} + \cdots + \frac{(-1)^n x^{2n}}{(2n+1)!} + \cdots$$

for all real numbers x.

(a) Find $f'(0)$ and $f''(0)$. Determine whether f has a local maximum, a local minimum, or neither at $x = 0$. Give a reason for your answer.

(c) Show that $y = f(x)$ is a solution to the differential equation $xy' + y = \cos x$.

a) $f'(x) = -\frac{2x}{3!} + \frac{4x^3}{5!} - \frac{6x^5}{7!} + \cdots + \frac{(-1)^n(2n)x^{2n-1}}{(2n+1)!}$

$f'(0) = -0 + 0 - 0 + \cdots 0$, $\boxed{f'(0) = 0}$

$f''(x) = \frac{-2}{3!} + \frac{12x^2}{5!} - \frac{30x^4}{7!} + \cdots \frac{(-1)^n(2n)(2n-1)x^{2n-2}}{(2n+1)!}$

$\boxed{f''(0) = -2/3!}$ Since $f'(0) = 0$ and $f''(0) < 0$, f has a critical value and is concave down. $\boxed{\text{Rel Max}}$

c) Build $xy' + y$ to show the expansion is $\cos x$

$y' = \frac{-2}{3!}x + \frac{4x^3}{5!} - \frac{6x^5}{7!} + \cdots + \frac{(-1)^n(2n)x^{2n-1}}{(2n+1)!}$ multiply by x

$xy' = \frac{-2}{3!}x^2 + \frac{4x^4}{5!} - \frac{6x^6}{7!} + \cdots + \frac{(-1)^n(2n)x^{2n}}{(2n+1)!}$

$+ \quad y = 1 - \frac{1x^2}{3!} + \frac{1x^4}{5!} - \frac{1x^6}{7!} + \cdots + \frac{(-1)^n(1)x^{2n}}{(2n+1)!}$

$xy' + y = 1 - \frac{3x^2}{3!} + \frac{5x^4}{5!} - \frac{7x^6}{7!} + \cdots + \frac{(-1)^n(2n+1)x^{2n}}{(2n+1)!}$

$$\boxed{xy' + y = 1 - x^2/2! + x^4/4! - x^6/6! + \cdots + \frac{(-1)^n x^{2n}}{(2n)!} = \cos x\,!}$$

Ex. 12: The function f has derivatives of all orders at $x = 0$, and the Maclaurin series for f is $\sum_{n=2}^{\infty} \frac{\ln n}{3^n n^3} x^n$. $= \frac{\ln 2}{9 \cdot 8} x^2 + \frac{\ln 3}{27 \cdot 27} x^3 + \frac{\ln 4}{81 \cdot 64} x^4 + \cdots$

(a) Find $f'(0)$ and $f^{(4)}(0)$.

(b) Does f have a relative minimum, a relative maximum, or neither at $x = 0$? Justify your answer.

(c) Using the ratio test, determine the interval of convergence of the Maclaurin series for f. Justify your answer.

a) There is no f' term, so $\boxed{f'(0) = 0}$

$f^{(4)}(0)$: 4th degree term is $\frac{f^{(4)}(0)}{4!} \cancel{x^4} = \frac{\ln 4}{81 \cdot 64} \cancel{x^4}$

So, $\boxed{f^{(4)}(0) = \frac{4! \cdot \ln 4}{81 \cdot 64}}$

b) Since $f'(0) = 0$ and $f''(0) = \frac{2! \cdot \ln 2}{9 \cdot 8} > 0$, then f has a $\boxed{\text{relative min}}$

c) $\lim_{n \to \infty} \left| \frac{\ln(n+1) x^{n+1}}{3^{n+1} (n+1)^3} \cdot \frac{3^n n^3}{\ln n \cdot x^n} \right| < 1$

$= \lim_{n \to \infty} \left| \frac{n^3 \ln(n+1)}{(n+1)^3 \ln n} \cdot \frac{x}{3} \right| < 1$, $\left|\frac{x}{3}\right| < 1$, $-1 < \frac{x}{3} < 1$, $-3 < x < 3$

as $n \to \infty$, this goes to 1

$\boxed{-3 \le x \le 3}$

$x = -3$: $\sum \frac{\ln n \cdot (-3)^n}{3^n \cdot n^3} = \sum \frac{(-1)^n \ln n}{n^3}$ i) $\lim_{n \to \infty} \frac{\ln n}{n^3} = 0$ ✓ ii) $\frac{\ln(n+1)}{(n+1)^3} < \frac{\ln n}{n^3}$ ✓

$\therefore$ $\boxed{\text{converges}}$ by alternating series test

$x = 3$: $\sum \frac{\ln n \, \cancel{3^n}}{\cancel{3^n} n^3} = \sum \frac{\ln n}{n^3} < \sum \frac{n}{n^3} = \sum \frac{1}{n^2}$ $\boxed{\text{converges}}$ by direct comparison to convergent $p = 2$ series

AP Calculus II
Notes 9.9
Geometric Power Series

Consider the series $\sum_{n=0}^{\infty} x^n$. We know that this sum can be written as:

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + x^4 + \ldots + x^n$$

This expression closely resembles the sum of a geometric series with $a =$ 1 and $r =$ x. Therefore,

the sum of the series would be: $\sum_{n=0}^{\infty} x^n = \frac{a_1}{1-r} = \frac{1}{1-x}$

We can now express functions that look like the sum of a geometric sequence as a **geometric power series** using the general representation that has a center at c:

$$\sum_{n=0}^{\infty} a(x-c)^n = a + a(x-c) + a(x-c)^2 + a(x-c)^3 + \ldots = \frac{a}{1-(x-c)}$$

Ex. 1: Write the following expressions as a geometric power series and then write the first 4 terms:

a) $\frac{12}{1-\frac{1}{4}} = \frac{12}{3/4} = 16$

$$\sum_{n=0}^{\infty} 12\left(\tfrac{1}{4}\right)^n = 16$$

$$= 12 + 3 + \frac{3}{4} + \frac{3}{16} + \ldots$$

$$= 16$$

b) $\frac{3}{1-(x+2)} = \frac{3}{-1-x}$

$$\sum_{n=0}^{\infty} 3(x+2)^n$$

$$= 3 + 3(x+2) + 3(x+2)^2 + 3(x+2)^3 + \ldots$$

c) $\frac{5x^2}{1+x} = \frac{5x^2}{1-(-x)}$

$$\sum_{n=0}^{\infty} 5x^2(-x)^n$$

$$= \sum_{n=0}^{\infty} (-1)^n 5x^{n+2}$$

$$= 5x^2 - 5x^3 + 5x^4 - 5x^5 + \ldots$$

need $a/(1-r)$ form!

Ex. 2: Find a power series for $f(x) = \dfrac{4}{x+5}$, centered at the following values. Then, find the interval of convergence for each power series.

a) $c = 0$

$$: f(x) = \frac{4}{5+x} = \frac{4/5}{1+x/5} = \frac{4/5}{1-(-x/5)}$$

Center at $c=0$ since just x.

$$\boxed{\frac{4}{x+5} = \sum_{n=0}^{\infty} \frac{4}{5}\left(\frac{-x}{5}\right)^n = \sum_{n=0}^{\infty} \frac{4}{5}(-1)^n\left(\frac{x}{5}\right)^n = \frac{4}{5} - \frac{4x}{25} + \frac{4x^2}{125} - \ldots}$$

This series (and being equal to $\frac{4}{x+5}$) converges only during its interval of convergence. Since this is geometric, the series converges when $|r|<1$. $r = x/5$ here

$\left|\frac{x}{5}\right| < 1$, $-1 < \frac{x}{5} < 1 \rightarrow \boxed{-5 < x < 5}$ * No need to test endpoints since $r=1$ diverges always *

b) $c = 3$

$$f = \frac{4}{5+x} \xrightarrow[\text{centered}=3]{\text{make}} \frac{4}{5+3+(x-3)} = \frac{4}{8+(x-3)} = \frac{4/8}{1+\frac{x-3}{8}} = \frac{1/2}{1-\left(\frac{x-3}{-8}\right)}$$

$$\boxed{\frac{4}{x+5} = \sum_{n=0}^{\infty} \frac{1}{2}\left(\frac{x-3}{-8}\right)^n = \sum_{n=0}^{\infty} \frac{1}{2}(-1)^n\left(\frac{x-3}{8}\right)^n = \frac{1}{2} - \frac{x-3}{16} + \frac{(x-3)^2}{128} - \ldots}$$

(absolute value takes care of $(-1)^n$)

IoC: $\left|\frac{x-3}{8}\right| < 1$

$-1 < \frac{x-3}{8} < 1 \rightarrow -8 < x-3 < 8 \rightarrow \boxed{-5 < x < 11}$

So, $\dfrac{4}{x+5} = \displaystyle\sum_{n=0}^{\infty} \frac{1}{2}(-1)^n\left(\frac{x-3}{8}\right)^n$ as long as x is between $-5 < x < 11$. Notice -5 is also a domain restriction... coincidence??

u-sub won't work ...no partial fractions, etc...

Ex. 3: Evaluate $\int \frac{6x}{1-x^3}dx$ using power series centered at 0.

But we can integrate a polynomial!

$$\frac{6x}{1-x^3} = \sum_{n=0}^{\infty} 6x(x^3)^n = \sum_{n=0}^{\infty} 6x^{3n+1} = 6x + 6x^4 + 6x^7 + 6x^{10} + \dots$$

$$\boxed{\int \frac{6x}{1-x^3}dx = \frac{6x^2}{2} + \frac{6x^5}{5} + \frac{6x^8}{8} + \frac{6x^{11}}{11} + \dots \frac{6x^{3n+2}}{3n+2} + C}$$

Ex. 4: If $h'(x) = 1 - 3x + 9x^2 - 27x^3 + \dots$

a) Write the function represented by $h'(x)$ and then write the Maclaurin series for $h(x)$.

h' has repeated multiplication of $-3x$ ..

$h' = \sum_{n=0}^{\infty} (-3x)^n$ This is geometric so $\boxed{h' = \frac{1}{1 - -3x}}$

$$h(x) = \int h'(x)dx = \int \sum_{n=0}^{\infty} (-1)^n 3^n x^n dx = \boxed{\sum_{n=0}^{\infty} \frac{(-1)^n 3^n x^{n+1}}{n+1} + C}$$

b) Find the radius of convergence of $h(x)$.

h(x) isn't geometric (h' is)

$$\lim_{n\to\infty} \left| \frac{3^{n+1} x^{n+2}}{n+2} \cdot \frac{n+1}{3^n x^{n+1}} \right| < 1 \rightarrow \lim_{n\to\infty} \left| \frac{n+1}{n+2} \cdot 3x \right| < 1$$

as $n \to \infty$, this goes to 1

$$|3x| < 1 \rightarrow -1 < 3x < 1, \quad -\frac{1}{3} < x < \frac{1}{3} \quad \text{so} \quad \boxed{R = 1/3}$$

$\leftarrow c = 0 \rightarrow$

$-1/3 \qquad +1/3$

Ex. 5:

The Maclaurin series for the function f is given by $f(x) = \sum_{n=2}^{\infty} \frac{(-1)^n (2x)^n}{n-1}$ on its interval of convergence.

(a) Find the interval of convergence for the Maclaurin series of f. Justify your answer.

(b) Show that $y = f(x)$ is a solution to the differential equation $xy' - y = \frac{4x^2}{1+2x}$ for $|x| < R$, where R is the radius of convergence from part (a).

a) $\lim_{n\to\infty} \left| \frac{(2x)^{n+1}}{n} \cdot \frac{n-1}{(2x)^n} \right| < 1 \rightarrow \lim_{n\to\infty} \left| \frac{n-1}{n} \cdot 2x \right| < 1$

as $n \to \infty$, this goes to 1

$|2x| < 1$, $-1 < 2x < 1$, $-1/2 < x < 1/2$ [not geometric, so test endpoints]

$x = -1/2$: $\sum_{n=2}^{\infty} \frac{(-1)^n(-1)^n}{n-1}$ $\boxed{\text{diverges}}$ by harmonic series

$x = 1/2$: $\sum_{n=2}^{\infty} \frac{(-1)^n \cancel{(1)^n}}{n-1}$ $\boxed{\text{converges}}$ by alternating harmonic

$$\boxed{-1/2 < x \le 1/2}$$

b) $y' = \sum_{n=2}^{\infty} \frac{(-1)^n n (2x)^{n-1} \cdot 2}{n-1}$ multiply by x

$(2x)^{n-1} \cdot (2x)'$

$xy' = \sum_{n=2}^{\infty} \frac{(-1)^n n (2x)^{n-1} \cdot 2x}{n-1} = \sum_{n=2}^{\infty} \frac{(-1)^n n (2x)^n}{n-1}$

$-y = \sum_{n=2}^{\infty} \frac{(-1)^n (2x)^n}{n-1}$

$xy' - y = \sum_{n=2}^{\infty} \frac{(-1)^n n (2x)^n}{n-1} - \sum_{n=2}^{\infty} \frac{(-1)^n 1 (2x)^n}{n-1} = \sum_{n=2}^{\infty} \frac{(-1)^n \cancel{(n-1)} (2x)^n}{\cancel{n-1}}$

$xy' - y = \sum_{n=2}^{\infty} (-1)^n (2x)^n = \sum_{n=2}^{\infty} (-2x)^n \xrightarrow{\text{geometric!}} \frac{(-2x)^2}{1 - -2x} = \boxed{\frac{4x^2}{1+2x}}$

AP Calculus II
Notes 9.7
Error Approximation

One of the bigger uses of Taylor polynomials is to approximate values of a transcendental/trigonometric function. This leads us to the idea of the accuracy of the approximation.

$$f(c) = \quad P_n(c) \quad + \quad R_n(c)$$

actual/exact = polynomial approximation + Remainder

So, $R_n(c) = f(c) - P_n(c)$. This means that the **error** of the approximation can be expressed as:

Error = $\left| f(c) - P_n(c) \right|$

error is considered positive even if above or below

We can find this error using one of two methods, which depend on whether the polynomial alternates. We will start off with if the polynomial does alternate:

Alternating Series Remainder:

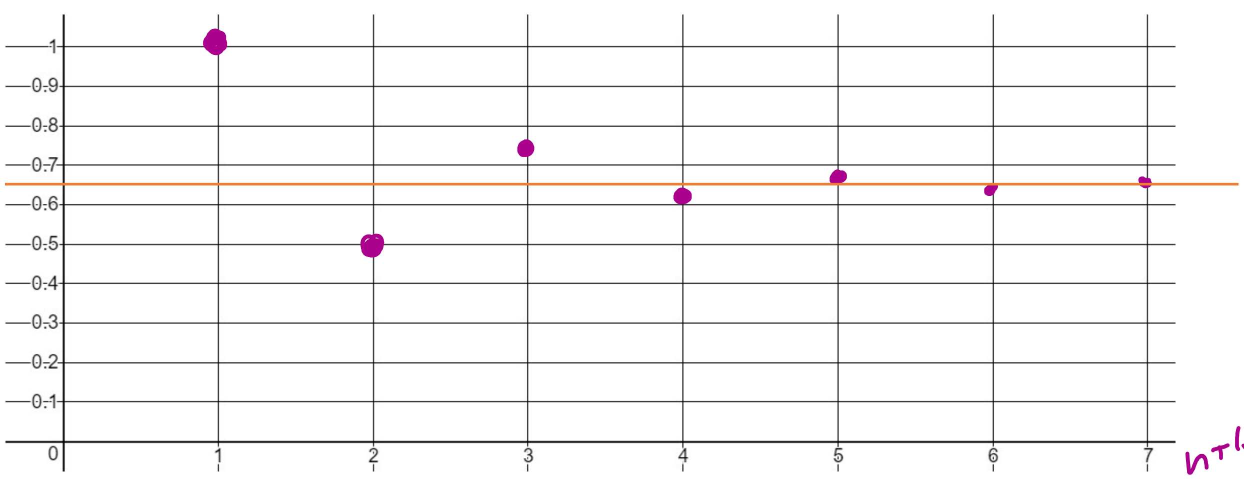
If a series $\sum_{n=1}^{\infty} a_n$ is **alternating, decreasing in absolute value** and $\lim_{n\to\infty} a_n = 0$, then the series converges.

If you look at the partial sums of the series, it will bounce back and forth around the actual sum. To model this, let's look at the following geometric series:

$$\sum_{n=0}^{\infty}\left(-\frac{1}{2}\right)^n = 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16} - \frac{1}{32} + \cdots$$

Knowing that this series is geometric, we know that this series adds up to $\frac{1}{1-(-\frac{1}{2})} = \frac{2}{2+1} = \boxed{\frac{2}{3}}$ actual value.

Let's imagine we didn't know what this series ***equals***, so let's ***approximate*** the sum using the first few partial sums.



$1 \qquad 1-\frac{1}{2} \qquad 1-\frac{1}{2}+\frac{1}{4} \qquad 1-\frac{1}{2}+\frac{1}{4}-\frac{1}{8} \qquad 1-\frac{1}{2}+\frac{1}{4}-\frac{1}{8}+\frac{1}{16}$

By introducing the next term in the series, you go from 1 estimate to the next estimate. You also pass through the actual value.

So, if one partial sum is larger than the true value, then subtracting the next term will cause the partial sum to be smaller and thus, less than the true value. Then, adding the next term will cause the partial sum to be larger and so on… The error is the difference between any of the partial sums and the actual value, but by adding (or subtracting) the next term, the partial sum will go past the actual value. Therefore, the **maximum error** is less than or equal to the **first (next) neglected term** in the series.

Again, this applies to **convergent alternating series that decrease in absolute value to 0.**

looks geometric!

Ex. 1: Given $g(x) = \frac{6}{2+x}$, write the first four nonzero terms for the Maclaurin polynomial and the general term for $g(x)$. Using this polynomial, approximate $g(1)$ and determine the maximum error.

$$g = \frac{6}{2+x} = \frac{3}{1+x/2} = \frac{3}{1-(-x/2)} = \sum_{n=0}^{\infty} 3\left(-\frac{x}{2}\right)^n = 3 - \frac{3x}{2} + \frac{3x^2}{4} - \frac{3x^3}{8} + \dots$$

$$\boxed{P_4(x) = 3 - \frac{3}{2}x + \frac{3}{4}x^2 - \frac{3}{8}x^3 + \frac{3}{16}x^4} \quad \boxed{g(1) \approx P_4(1) = 1.9125}$$

Since g is a convergent alternating series that decreases in absolute value to 0…

$$|g(1) - P_4(1)| \le \left|\begin{matrix}\text{next}\\ \text{term}\end{matrix}\right| \to |g(1) - 1.9125| \le \left|\frac{-3x^5}{32}\right|_{x=1} \Rightarrow \boxed{0.09375}$$

$\boxed{g(1) = \frac{6}{3} = 2 \quad \text{Actual error} = 0.0875}$

Ex. 2: Given $f(x) = \sin x$, write the fifth degree Maclaurin polynomial and the general term for $f(x)$. Using this polynomial, approximate $\sin(0.1)$ and determine the maximum error.

$$\sin x \approx x - \frac{x^3}{3!} + \frac{x^5}{5!} = P_5(x), \quad \text{General term: } \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

$$\sin(0.1) \approx P_5(0.1) = 0.0998334166667$$

Since f is a convergent, alternating series that decreases in absolute value to 0,

$$|\underset{\text{actual}}{\sin 0.1} - \underset{\text{approximate}}{P_5(0.1)}| \le \left|-\frac{x^7}{7!}\right|_{x=0.1}$$

$\boxed{\text{max error} = 1.9841 \times 10^{-11}}$

$[\sin 0.1 = 0.09983341664]$ ← 11th digit

Ex. 3: Given $f(x) = e^{-2x}$,

a) Write the third degree Maclaurin polynomial $P_3(x)$ for $f(x)$.

either build F or develop from general form of poly.

$e^x = 1 + x + x^2/2! + x^3/3! + \ldots$ replace x with -2x

$$\boxed{e^{-2x} \approx P_3(x) = 1 - 2x + \frac{4x^2}{2!} - \frac{8x^3}{3!}}$$

b) Using $P_3(x)$, approximate $f(0.5)$. Then, show that $f(0.5)$ differs from $P_3(0.5)$ by less than $\frac{1}{20}$.

$$f(0.5) = e^{-1} \approx P_3(0.5) = 1 - 1 + \frac{1}{2} - \frac{1}{6} = \boxed{0.3333333}$$

since f is a convergent, alternating series that decreases in absolute value to 0

$$\left|e^{-1} - 0.333\right| \leq \left|\frac{16(0.5)^4}{4!}\right| = \frac{16\left(\frac{1}{2}\right)^4}{4!} = \frac{16\left(\frac{1}{16}\right)}{4!} = \boxed{\frac{1}{24} < \frac{1}{20}} ✓$$

Ex. 4: The Taylor series for f about $x = 2$ converges to f for $-1 \leq x \leq 5$. The nth-degree Taylor polynomial for f about $x = 2$ is given by $P_n(x) = \sum_{k=1}^{n} (-1)^k \frac{(x-2)^k}{3^k(k^2+1)}$. What is the smallest degree n of the Taylor polynomial that guarantees that $|f(3) - P_n(3)| \leq \frac{1}{100{,}000}$?

Since f is a convergent, alternating series that decreases in absolute value to 0, the max error is $\frac{1}{100{,}000}$, which is the value of the next term.

if $n=4 \to 5^{th}$ degree term is max error $\to \frac{(3-2)^5}{3^5(25+1)} = \frac{1}{6318}$ ... not $\leq \frac{1}{100{,}000}$

if $n=6 \to 7^{th}$ degree term is max error $\to \frac{(3-2)^7}{3^7(49+1)} = \frac{1}{109350} \leq \frac{1}{100{,}000}$

if $n=5 \to 6^{th}$ degree term is max error $\to \frac{(3-2)^6}{3^6(36+1)} = \frac{1}{26{,}973}$

$\boxed{\text{So } n=6}$

Ex. 5: The Taylor series about $x = 4$ for a certain function f converges to $f(x)$ for all x in the interval of convergence. The nth derivative of f at $x = 4$ is given by

$$f^{(n)}(4) = \frac{(-1)^n (n+1)!}{2^n} \quad \text{and } f(4) = 1$$

a) Write the third-degree Taylor polynomial for $f(x)$ centered at $x = 4$ and find the general term.

Can't build from known series ...

$$P_3(x) = f(4) + \frac{f'(4)}{1!}(x-4) + \frac{f''(4)}{2!}(x-4)^2 + \frac{f'''(4)}{3!}(x-4)^3$$

$f(4) = 1$

$f'(4) = \frac{-2!}{2}$

$f''(4) = 3!/4$

$f'''(4) = -4!/8$

$$P_3(x) = 1 - \frac{2!/2}{1!}(x-4) + \frac{3!/4}{2!}(x-4)^2 - \frac{4!/8}{3!}(x-4)^3$$

$$\boxed{P_3(x) = 1 - \frac{2}{2}(x-4) + \frac{3}{4}(x-4)^2 - \frac{4}{8}(x-4)^3}$$

general term $= \frac{(-1)^n(n+1)!}{2^n} / n! = \boxed{\frac{(-1)^n(n+1)}{2^n}(x-4)^n}$

b) Find the coefficient of the 10^{th} degree term of the Taylor series for f centered at $c = 4$.

if general term is $\frac{(-1)^n(n+1)}{2^n}(x-4)^n$

$n = 10 \rightarrow +\frac{11}{2^{10}}(x-4)^{10}$ so $\boxed{11/2^{10}}$

c) Show that the third-degree Taylor polynomial approximates $f(5)$ with an error less than $\frac{1}{3}$.

Since f is a convergent, alternating series that decreases in absolute value to 0,

$$|f(5) - P_3(5)| \le \left| \begin{matrix}4^{\text{th}} \text{ degree} \\ \text{term at } x=5\end{matrix} \right| \rightarrow \frac{5}{16}(5-4)^3 = \boxed{\frac{5}{16} < \frac{1}{3}}$$

yes ✓

What if the series is not alternating?

If the series is not alternating, you must use the Lagrange Error in which the maximum error will occur at some value z, such that z is in between x and c.

center $\to c \le z \le x \leftarrow$ value used to find approximate

Lagrange Error Bound/Taylor's Theorem:

For $f(x) = f(c) + \frac{f'(c)}{1!}(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \frac{f'''(c)}{3!}(x-c)^3 + \cdots + \frac{f^{(n)}(c)}{n!}(x-c)^n + R_n(x)$

Still uses next term

where the maximum value of the remainder is $R_n(x) = \left|\frac{f^{(n+1)}(z)}{(n+1)!}(x-c)^{n+1}\right|$.

This is very similar to the Alternating Series Remainder in that they both use the next term in the series. However when using Lagrange Error, you must find or use reasoning to determine the z value that will make the next derivative the maximum value possible. This will make the remainder the largest possible and will truly represent the maximum error.

Ex. 6: Given the function $f(x) = 3e^x$,

a) Write the fourth-degree Maclaurin polynomial for $f(x)$ and the general term. Use this fourth-degree polynomial to approximate $3e^2$.

$f(x) \approx P_4(x) = 3 + 3x + \frac{3x^2}{2!} + \frac{3x^3}{3!} + \frac{3x^4}{4!}$ general term: $\frac{3x^n}{n!}$

$\boxed{f(2) = 3e^2 \approx P_4(2) = 21}$

b) Find the maximum error of this approximation to the actual value of $3e^2$.

f is not alternating, so must use Lagrange

$\left| f(2) - 21 \right| \le \left| \frac{f^{(5)}(z)}{5!} x^5 \right|_{x=2}$ where z is a number between $0 \le z \le 2$ that will make $f^{(5)}(x)$ a maximum

actual approximate

$f^{(5)}(x) = 3e^x$, which is always increasing. So therefore, $z = 2$ would make $f^{(5)}(x)$ a max.

max error $= \frac{f^{(5)}(2)}{5!}(2)^5 = \boxed{\frac{3e^2}{5!}(2)^5 = 5.9112}$

Ex. 7: The function $f(x) = \frac{1}{(1-x)^2}$ can be expressed as the series $\sum_{n=1}^{\infty} nx^{n-1}$ for all values of x within its interval of convergence.

a) Find the interval of convergence of $f(x)$.

$$\lim_{n\to\infty}\left|\frac{(n+1)x^n}{nx^{n-1}}\right|<1 \rightarrow \lim_{n\to\infty}\left|\frac{n+1}{n}\cdot x\right|<1$$

as $n\to\infty$, this goes to 1

$|x|<1$

$-1<x<1$

check endpoints!

$x=-1$: $\sum n(-1)^{n-1}$ diverges by n^{th} term test for divergence

$x=1$: $\sum n$ diverges by n^{th} term test for divergence

$\boxed{(-1,1)}$

b) Use the first 4 nonzero terms of f to approximate the value of $f\left(\frac{1}{3}\right)$. Show that this approximation differs from $f\left(\frac{1}{3}\right)$ by less than $\frac{3}{4}$.

$$f \approx 1 + 2x + 3x^2 + 4x^3 = P_3(x), \quad \boxed{P_3(1/3) = 2.148148}$$

not alternating. So must maximize the $f^{(4)}$ & next derivative

$f = (1-x)^{-2}$

$f' = +2(1-x)^{-3}$

$f'' = 6(1-x)^{-4}$

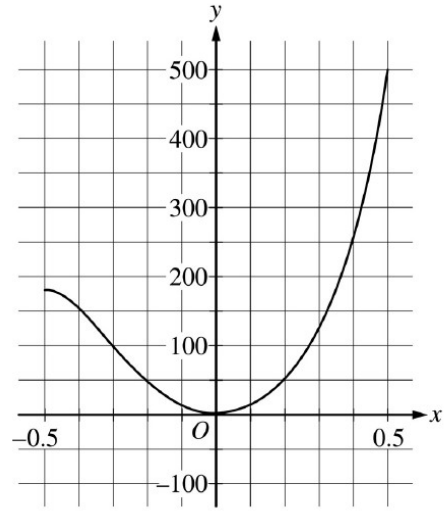
$f''' = 24(1-x)^{-5}$

$f^{(4)} = 120(1-x)^{-6}$

So $f^{(4)}(x) = \frac{120}{(1-x)^6}$. So what value of z, where $0 \le z \le 1/3$, will make $f^{(4)}$ the greatest value. As x increases, the denominator gets smaller. This cause the $f^{(4)}$ to get larger ... so $z = 1/3$ is the max!

$$\left|f(1/3) - P_3(1/3)\right| \le \left|\frac{f^{(4)}(1/3)}{4!}\left(\frac{1}{3}\right)^4\right| \rightarrow \frac{\frac{120}{(1-1/3)^6}}{4!}\left(\frac{1}{3}\right)^4 = \boxed{0.703 < \frac{3}{4}}$$

Ex. 8: Let the function f be defined as $f(x) = x^2 e^{x^2}$. The graph of $f^{(7)}$, the seventh derivative of f, can be seen below.



a) Write the first four nonzero terms and general term for the Maclaurin series for e^x. Then, use this series to write the first four nonzero terms and general term for the Maclaurin series for f.

$$e^x = 1 + x + x^2/2! + x^3/3! + \dots + x^n/n!$$

replace x with x^2

$$e^{x^2} = 1 + x^2 + x^4/2! + x^6/3! + \dots + x^{2n}/n!$$

multiply by x^2

$$\boxed{f = x^2 e^{x^2} = x^2 + x^4 + x^6/2! + x^8/4! + \dots + \frac{x^{2n+2}}{n!}}$$

b) Let $P_6(x)$ be the sixth-degree Taylor polynomial for f about $x = 0$. Find $P_6\left(\frac{1}{2}\right)$, then use the Lagrange error bound along with the graph to find an upper bound on $\left|P_6\left(\frac{1}{2}\right) - f\left(\frac{1}{2}\right)\right|$.

$$P_6 = x^2 + x^4 + \frac{x^6}{2!} \rightarrow \boxed{P_6(1/2) = 0.3203125}$$

Since f is not alternating, use Lagrange!

$$|f(1/2) - P_6(1/2)| \le \left|\frac{f^{(7)}(z)}{7!} x^7\right|$$

$0 \le z \le 1/2$

- According to graph of $f^{(7)}$, the max of $f^{(7)}$ from $[0, 1/2]$ is 500

So, Max error $= \boxed{\frac{500}{7!}(1/2)^7 = 0.000775}$